QBEAST AI

Circuit and Liquidity Flags

Final calibrated submission guide | Release 2.4.4 | 20 October 2026

Purpose

Causal daily flags identify sessions when a printed price may not be practically tradable. The pipeline combines circuit, STEP, chronic illiquidity, and market-wide breaker reasons into a single trade gate.

Final rules

Stock rule	Tag condition	Release
Circuit staircase	5 consecutive same-direction limit-locked sessions, qualified by thin volume. If volume is missing, price evidence decides.	3 consecutive normal sessions
STEP staircase	3 unchanged-price sessions with no real trading: volume below 20% of normal, zero volume, or one price all day; OR 2 flat-then-2%+ jumps within 40 sessions.	2 normal sessions
Chronic illiquidity	Trailing 60 sessions: single-price >=20%, no-trade >=20%, or circuit days >=30%.	All measures below 10%, 10%, and 15%, respectively
Market-wide breaker	Breaker day.	Next day

ETF rule	Tag condition	Release
Circuit staircase	Same 5-day rule as stocks; not expected for ETFs.	3 consecutive normal sessions
STEP staircase	Same pattern rules as stocks; stale-volume cutoff is below 10% of normal.	2 normal sessions
Chronic illiquidity	Same 60-session test as stocks.	All measures below 10%, 10%, and 15%, respectively
Minimum traded value	From 1-Jan-2020: typical daily value below INR 250,000.	Back above INR 375,000
Market-wide breaker	Breaker day.	Next day

BLOCK MEANS: no buy, no sell, no exit while tagged. A holding remains held until the tag clears. Multiple active reasons are carried together.

Operational definitions

The final rule sheet leaves some terms operationally open. The shipped profile resolves them as follows:

Circuit staircase: thin volume

The calibrated circuit-volume test is the trailing 3-session average of real volume below 50% of the median of the prior 20 clean volume observations. This preserves the existing pipeline's calibrated definition. If volume is missing or unreliable, the causal OHLC fallback is used and a warning is emitted.

STEP staircase

A stale session has an unchanged close versus the prior close and no real trading: zero volume, volume below the asset-class cutoff, or a single-price bar. The stock cutoff is 20% of its volume baseline; the ETF cutoff is 10%. A STEP event is the first non-stale session after at least 2 stale sessions with an absolute move of at least 2%. The second such event within 40 sessions tags the pattern. Three consecutive stale sessions also tag it.

Chronic illiquidity

Ratios use a trailing 60-trading-session window, with at least 20 observations before a new tag. Release uses hysteresis: all three measures must be strictly below their release cutoffs. A missing-volume session is unknown, not automatically a no-trade session; zero-volume and gap-filled sessions follow the configured volume-quality rules.

ETF traded-value floor

The policy says typical daily traded value. This release defines typical as the median of the trailing 20 real-volume observations of close multiplied by volume. The rolling history can include valid pre-2020 observations for warm-up, but the floor cannot activate before 1-Jan-2020. The floor enters below INR 250,000 and stays active until the median reaches INR 375,000. It is disabled for stocks.

Causality and time handling

Each date's flags use only information available through that date. Windows count trading sessions, not calendar days. The market-wide breaker reason applies on the breaker session; the daily gate does not carry that reason to the following session.

Review this interpretation with the rule owner if 'typical traded value' is intended to mean a different statistic or rolling window.

Integration and validation

Pipeline files

File	Role
circuit_liquidity_flags.py	Causal flag logic, chronic state, gate, batch and streaming interfaces.
qbeast_flags_config.json	Shipped stock and ETF profiles, thresholds, and input/output paths.
flags_io.py / run_four_flags.py	Data loading and stock/ETF batch outputs.
daily_backtest_loop.py	Streaming daily loop and tracker-state persistence.
verify_install.py	Synthetic end-to-end release check; requires no market history.

Local verification

Unit functions: 63/63 passed. Built-in installation verifier: 62/62 passed, including synthetic stock and ETF processing, causal checks, batch/streaming parity, and the calibrated ETF floor settings.

Data-validation boundary

No full-history ETF result is claimed. The supplied workspace does not contain ETF EOD files in the configured QBEAST data layout. Run the pipeline against the submission's complete stock and ETF inputs before citing historical flag counts or impact estimates.

Submission checklist

Check	Status / action
Rule profile	Implemented in the shipped JSON configuration and documented here.
Synthetic validation	Passed: 63 unit checks and 62 end-to-end install checks.
Full-data run	Run `python run_four_flags.py --config qbeast_flags_config.json --data <Final_Data>` on the actual input tree.
Release package	Include the source/config files, README, SPEC, CHANGELOG, this guide, and requirements.

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